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Nonstationary patterns in demographic traits covary with Chinook salmon marine distributions

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Abstract

The abundance of many Chinook salmon (*Oncorhynchus tshawytscha*) stocks has declined despite reductions in harvest. We used state-space models parameterized with data from 57 Chinook salmon indicator stocks, ranging from coastal Oregon to southeast Alaska, to quantify long-term (since 1972 release year) changes in juvenile marine survival rate and mean age-at-maturity, as well as identify stock groupings with coherent dynamics. We found that juvenile marine distribution—rather than freshwater life history, run timing, or adult marine distribution—was the best predictor of trends in both survival and age. Only subyearling stocks that enter the Strait of Georgia showed evidence of transitioning to a low juvenile survival period, other groupings exhibited low and stable or cyclical patterns in survival. Conversely, declines in mean age-at-maturity were widespread and do not appear to have stabilized, suggesting that future declines in Chinook salmon population productivity may be influenced by earlier maturation or increased adult mortality, rather than further reductions in juvenile marine survival.

Key words: age-at-maturity, dynamic factor analysis, life history, population productivity, population demography, spatial distribution, survival

Résumé

L'abondance de nombreux stocks de saumons chinooks (*Oncorhynchus tshawytscha*) a diminué malgré des réductions des prises. Nous utilisons des modèles d'espaces d'états paramétrés avec des données sur 57 stocks indicateurs de saumons chinooks allant du littoral de l'Oregon au sud-est de l'Alaska pour quantifier les changements à long terme (depuis l'année de lâcher 1972) des taux de survie en mer de juvéniles et de l'âge moyen à la maturité, ainsi que pour cerner des groupes de stocks ayant une dynamique cohérente. Nous constatons que la répartition en mer des juvéniles, plutôt que le cycle biologique en eau douce, le moment de la montaison ou la répartition en mer des adultes, est le meilleur prédicteur des tendances tant de la survie que de l'âge. Seuls les stocks d'individus de moins d'un an qui entrent dans le détroit de Georgia présentent des signes d'une transition vers une période de faible survie des juvéniles, les autres groupes présentant des motifs de survie faibles et stables ou cycliques. En revanche, des baisses de l'âge moyen à la maturité sont répandues et ne semblent pas se stabiliser, ce qui indiquerait que les baisses futures de la productivité de populations de saumons chinooks pourraient être influencées par une maturation plus précoce ou une mortalité accrue des adultes, plutôt que par d'autres baisses de la survie en mer de juvéniles. [Traduit par la Rédaction]

Mots-clés : âge à la maturité, analyse factorielle dynamique, cycle biologique, productivité des populations, démographie des populations, répartition spatiale, survie

Introduction

The future of many North American Chinook salmon (*Oncorhynchus tshawytscha*) stocks appears to be increasingly uncertain due to factors including poor marine survival (Ruff et al. 2017; Welch et al. 2021), increasingly synchronous dynamics (Kilduff et al. 2014; Dorner et al. 2017), and declines in age-at-maturity and body size (Ohlberger et al. 2018; Oke et al. 2020; Xu et al. 2020). Such changes may negatively impact ecosystems via reduced prey availability for predator populations (Ward et al. 2009; Ford et al. 2010) and by limiting nutrient flows between marine and terrestrial communities (Hocking and Reynolds 2011; Oke et al. 2020). Extended periods of low Chinook salmon productivity (i.e., the number of adult offspring produced by a given number of spawners) have also resulted in widespread harvest restrictions with substantial socio-economic costs to communities that depend on First Nations, commercial, or recreational harvest opportunities (Lindley et al. 2009;

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ADF&G Chinook Salmon Research Team 2013; Doutaz et al. 2021).

Low Chinook salmon population productivity is common in populations that spawn in areas with substantial anthropogenic disturbance, as well as those in relatively less impacted regions, and has been associated with poor juvenile marine survival (Welch et al. 2021). These characteristics suggest that large-scale environmental processes may act as bottlenecks to recovery, particularly during marine life history stages. Hypothesized mechanisms include bottom-up processes moderating growth, such as basin-scale environmental conditions and density-dependent interactions with abundant hatchery-origin chum (*Oncorhynchus keta*) and pink salmon (*Oncorhynchus gorbuscha*) (Mantua et al. 1997; Kilduff et al. 2014; Cunningham et al. 2018; Oke et al. 2020; Sobocinski et al. 2021). Top-down processes, namely predation, may also constrain juvenile survival (Chasco et al. 2017*a*; Nelson et al. 2018). A dominant paradigm within salmon ecology is that population productivity, after accounting for spawner abundance, is predominantly determined by juvenile survival (Beamish and Mahnken 2001). However, a growing body of evidence indicates that late-stage mortality rates may be relatively higher than previously considered (Chasco et al. 2017*a*; Seitz et al. 2019; Wargo Rub et al. 2019). These factors may underpin changes in age-at-maturity and body size (Manishin et al. 2021).

Efforts to identify mechanisms driving declines in productivity across the geographic range of Chinook salmon are complicated by the species' diverse life history, which includes not only a gradient of freshwater rearing strategies, but also population-specific marine distributions (Quinn 2018; Riddell et al. 2018). Marine spatio-temporal distributions differ among watersheds, as well as among populations within a watershed (Weitkamp 2010; Shelton et al. 2019; Freshwater et al. 2021). For example, populations with subyearling (ocean-type) life histories typically rear within 100–200 km of their natal estuary for their first marine year, while yearling (stream-type) life histories may exhibit a similar near-shore distribution or migrate rapidly offshore shortly after ocean entry (Fisher et al. 2007; Trudel et al. 2009; Tucker et al. 2011). After this first marine year, individuals from certain populations remain resident within near-shore areas (Chamberlin et al. 2011; Freshwater et al. 2021), while others undergo extensive continental shelf or offshore migrations (Weitkamp 2010; Shelton et al. 2019). Such differences determine a population's exposure to prey, competitors, predators, and fisheries, moderating growth and survival.

Identifying coherent trends in survival and mean age-at-maturity among populations provides information on the spatial-scale of mechanisms regulating population dynamics. Shared trends can be used to test for evidence of regime shifts or nonstationarity, which may limit the efficacy of management efforts to rebuild depleted populations. Previous analyses have used dimension reduction techniques to identify common trends in Chinook salmon productivity (Ohlberger et al. 2016; Dorner et al. 2017), juvenile survival (Ruff et al. 2017), and age-at-maturity (Ohlberger et al. 2018).

Here we update and expand on this work by estimating shared trends in juvenile survival rates and mean age-at-maturity among 57 stocks of Chinook salmon using quantitative techniques well-suited to "gappy" time series. Our analysis explicitly accounts for changes in harvest, which may confound patterns in age-at-maturity. Additionally, we evaluate the extent to which trends in these demographic traits covary with a suite of stock traits (release age, run timing, juvenile marine distribution, and adult marine distribution). We use these analyses to explore three related questions. First, do the same ecological traits covary with both trends in juvenile survival rate and mean age-at-maturity? Second, how do temporal patterns in juvenile survival and mean age-at-maturity differ among stocks grouped by these traits? Finally, how do patterns in juvenile survival and mean age-at-maturity differ from one another and what are the implications for Chinook salmon management?

Methods

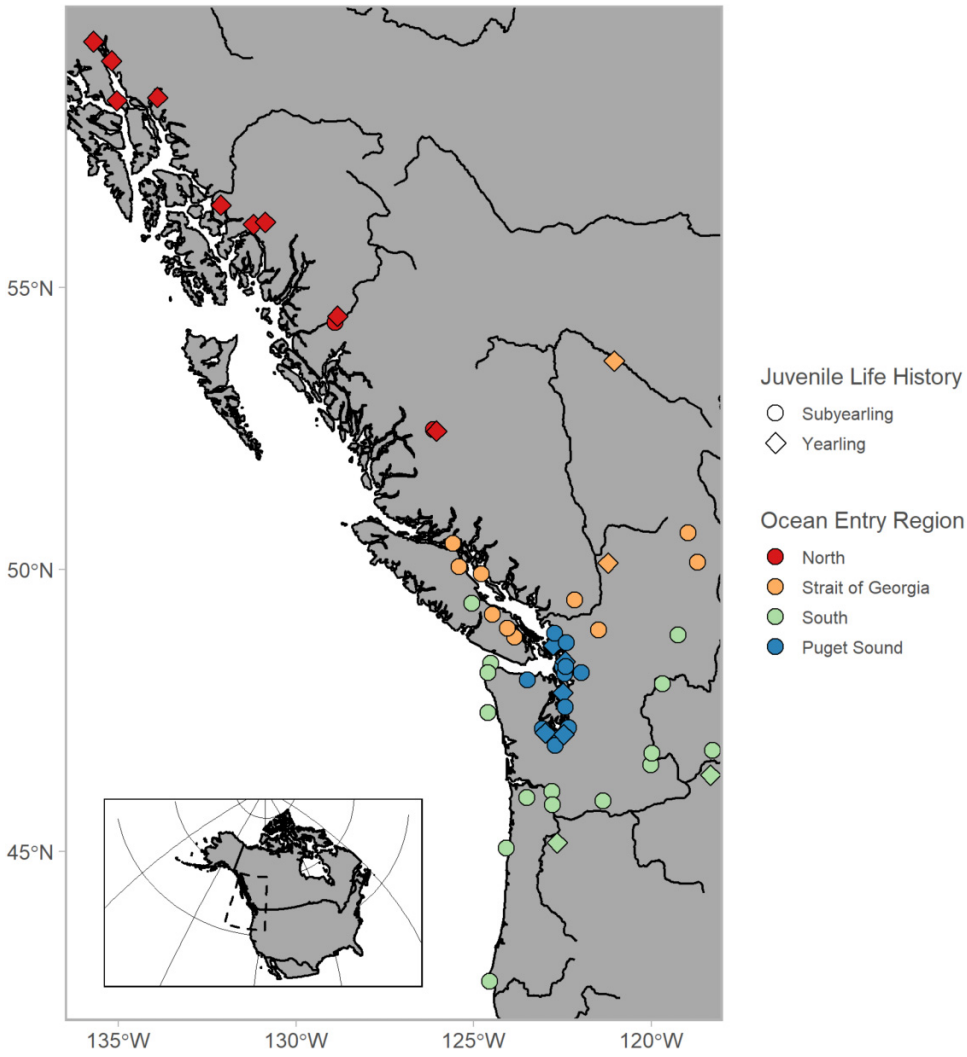
Study species

Our analysis focused on Chinook salmon populations used as exploitation rate indicator stocks (hereafter indicator stocks) by the Pacific Salmon Commission Chinook Technical Committee (CTC), a bi-national working group that assesses the status of Chinook salmon stocks managed under the Pacific Salmon Treaty (CTC 2021*a*). Several types of population statistics, including survival and maturation rate, are inferred from recoveries of coded wire tag (CWT) release groups (i.e., juvenile cohorts implanted with a CWT denoting stock identity and release date) (Bernard and Clark 1996; Johnson 2004). In this analysis, we examined 57 stocks with ocean entry locations distributed from coastal Oregon to southeast Alaska (Fig. 1). The stocks are predominantly of hatchery origin and include two release strategies (subyearling and yearling smolts), as well as three run timings (i.e., adult migration; spring, summer, fall; Table S1). The ocean entry years considered span more than four decades (1972–2017), although time series length differs among stocks (Table S1).

Estimates of juvenile marine survival and mean age-at-maturity

To explore trends in marine survival of smolts to fishery recruitment (hereafter juvenile marine survival) and mean age-at-maturity, we used data generated by a cohort reconstruction performed as part of the CTC's Exploitation Rate Analysis (CTC 2021*a*). Briefly, the abundance of a cohort within a stock is estimated using CWT recoveries (expanded for sampling rate) in preterminal (marine) and terminal (freshwater) fisheries, as well as on the spawning grounds. Since Chinook salmon age-at-maturity varies within a stock and, unlike many other Pacific salmon species, Chinook salmon recruit to fisheries before maturing, the CTC cohort reconstruction must account for age-specific harvest, natural mortality, and variable age-at-maturity. We note that juvenile marine survival and mean age-at-maturity are model-derived

Fig. 1. Map of the study area showing approximate freshwater rearing locations. Symbols represent freshwater release age and colours represent juvenile ocean entry regions associated with the top-ranked model (Table 2). Map projection is WGS84. Made with shapefiles from Natural Earth (naturalearthdata.com). [Colour online.]



quantities, but are assumed to be estimated without error for the purposes of this analysis.

Cohort abundance N of stock i and t_b (brood year) at age a (relative to the brood year) is estimated as

$$N_{i,t_b,a} = C_{i,t_b,a} + S_{i,t_b,a} + N^{\star}_{i,t_b,a+1} \quad (1)$$

where C is catch in all fisheries (including an adjustment for incidental mortality), S is escapement, and $N^{\star}_{a+1}$ is cohort abundance of the next oldest age class before accounting for natural mortality. Since $N^{\star}_{a+1} = 0$ for a stock's maximum age A, the analysis is performed sequentially from the oldest age class.

$N^{\star}_a$ is a function of N_a and the annual, age-specific natural mortality rate m_a:

$$N^{\star}_{i,t_b,a} = \frac{N_{i,t_b,a}}{1 - m_a} \quad (2)$$

Age-specific maturation rate γ_a is calculated as

$$\gamma_{i,t_b,a} = \frac{C^{\star}_{i,t_b,a} + S_{i,t_b,a}}{N_{i,t_b,a}} \quad (3)$$

where $C^{\star}$ is catch in terminal areas, as defined by CTC (2021*a*).

Note that m_a and γ_a cannot be estimated independently. To account for this, the CTC fixes natural mortality for each age class, i.e., $\mathbf{m} = (m_2, m_3, m_4, m_5, \cdots, m_A) = (0.4, 0.3, 0.2, 0.1, \cdots, 0.1)$ (CTC 2021*a*). A key implication of this assumption is that interannual and interstock variability in natural mortality will be reflected as changes in maturation rate. Once the cohort reconstruction is complete, spawner abundance in the absence of fishing $S^{\star}_{i,t_b,a}$ is calculated as the product of abundance and maturation rate ($N_{i,t_b,a}\gamma_{i,t_b,a}$).

Juvenile marine survival J is calculated as the ratio of estimated cohort size at the age of recruitment to the fishery a_f

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to cohort size at the age of release a_r:

$$(4) \quad J_{i,t_b} = \frac{N_{i,t_b,a_f}}{N_{i,t_b,a_r}}$$

where ($a_f = 2$, $a_r = 1$) for subyearling stocks and ($a_f = 3$, $a_r = 2$) for yearling stocks.

Changes in fishery practices, particularly the harvest of immature individuals and other changes in selectivity, may confound patterns in age-at-maturity inferred from raw CWT recoveries. Thus, we calculated mean age-at-maturity $K^{\star}_{i,t_b}$ in the absence of fishing:

$$(5) \quad K^{\star}_{i,t_b} = \frac{\sum_{a=a_f}^{A} S^{\star}_{i,t_b,a} a}{S^{\star}_{i,t_b}}$$

Chinook salmon stock groupings

We created stock groupings of varying complexity based on release age (yearling or subyearling), run timing (spring, summer, or fall), juvenile marine distribution, and adult marine distribution. For the purposes of this analysis we define juvenile Chinook salmon as individuals that have not yet reached a size where they can be legally retained in marine fisheries (generally some time in their second ocean year), while adults have recruited to the fishery. Juvenile marine distribution groupings reflect either spawning regions (a proxy for ocean entry locations) or spawning region and life history strategy, given evidence of differences between yearling and subyearling distributions for some stocks (Fisher et al. 2007; Tucker et al. 2011). Conversely, groupings based on adult distributions are derived from analyses of CWT recoveries in fisheries (Weitkamp 2010; Shelton et al. 2019; CTC 2021*a*), which indicate marine distributions after recruitment are broadly related to spawning region, but contain additional variation among stocks (Table 1). For example, some east coast Vancouver Island stocks have a local distribution, while others are northern migrants. For both juvenile and adult distributions, we also considered a suite of simpler models that grouped stocks at larger spatial scales. We recognize that these groupings are imperfect proxies (e.g., juvenile distributions may also represent adult distributions immediately prior to freshwater spawning migrations, distributions based on fishery recoveries provide little information where fisheries are absent). Nevertheless, they provide qualitative information that can be used to guide model structure. A full list of stocks and their associated traits is provided in Table S1.

Modeling framework

We first estimated state-space models in a maximum likelihood framework and used model selection to identify stock groupings (i.e., release age, run timing, juvenile distribution, or adult distribution) with the greatest evidence of underlying shared dynamics. We next estimated coherent trends (within each of the most parsimonious stock groupings identified via model selection) using dynamic factor analysis (DFA) implemented in a Bayesian framework. This is akin to the strategy utilized by Ruff et al. (2017) to explore patterns in juvenile Chinook salmon marine survival, except our analysis included a larger number of stocks over a longer number of years, fit DFAs using Bayesian techniques for increased flexibility, and incorporated two response variables–juvenile survival rate and mean age-at-maturity.

We logit-transformed survival rate data, i.e., logit(J), to meet assumptions of normally distributed error when fitting models. We aligned all time series by release year t_r (i.e., where t_r is $t_b + 1$ and $t_b + 2$ for subyearling and yearling stocks, respectively).

Observed time series of stock-specific survival rates or mean age-at-maturity were modeled as a linear combination of multiple latent (i.e., hidden) trends:

$$(6) \quad \boldsymbol{y}_{t_r} = \mathbf{Z}\boldsymbol{x}_{t_r} + \boldsymbol{\alpha} + \boldsymbol{v}_{t_r} \quad \text{where } \boldsymbol{v}_{t_r} \sim \text{MVN}(0, \mathbf{R})$$

where $\boldsymbol{y}$ represents a vector with length h stocks of the response variable (i.e., juvenile survival rates or mean age) at time t_r (release year), $\mathbf{Z}$ is a matrix of dummy variables, $\boldsymbol{x}$ is a vector of hidden trends with length g stock groups, and $\boldsymbol{\alpha}$ is a vector of time series-specific scaling parameters. $\boldsymbol{v}$ represents residual error which is assumed to be multivariate normally distributed with variance-covariance matrix $\mathbf{R}$. We constrained $\mathbf{R}$ to a single, shared variance term and zero covariance (see Dynamic factor analysis section for justification; models with alternative structures for $\mathbf{R}$ had similar performance).

We modelled hidden trends as random walks:

$$(7) \quad \boldsymbol{x}_{t_r} = \boldsymbol{x}_{t_r-1} + \boldsymbol{u} + \boldsymbol{w}_{t_r} \quad \text{where } \boldsymbol{w}_{t_r} \sim \text{MVN}(0, \mathbf{Q})$$

where $\boldsymbol{\mu}$ is a vector of group-specific intercepts allowing for divergent trends. Process error $\boldsymbol{w}$ is also assumed to be multivariate normally distributed with variance-covariance matrix $\mathbf{Q}$, which we modeled as unconstrained (i.e., group-specific variance estimates and nonzero covariance).

We evaluated specific hypotheses about the number of underlying processes and which stocks those processes applied to by manipulating $\mathbf{Z}$. For example, the simplest model assumed two latent processes, one that applied to yearling stocks and the other to subyearling stocks. The most complex model included nine latent processes, representing distinct juvenile distributions at relatively fine spatial scales and the two freshwater life history strategies (there are no yearling OR/WA/WCVI coastal indicator stocks). Thus, g varied from two to nine among models. We identified the most parsimonious stock groupings for each response variable by comparing models using AIC_c. We conducted this portion of the analysis in a maximum likelihood framework using the MARSS package (Holmes et al. 2014).

We next used DFA to estimate latent trends within each stock grouping. DFA is a dimension reduction technique, analogous to principal component analysis, where g latent trends shared among n time series are modeled as temporally autocorrelated processes (Zuur et al. 2003; Holmes et al. 2014). The dynamic factor model is similar to the model described above (eqs. 6 and 7), but includes several modifications. Instead of the components of $\mathbf{Z}$ being fixed at 0 or 1, they are estimated and represent factor loadings on the shared trends. The process model intercepts $\boldsymbol{\mu}$ and

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Table 1. Ecological traits used to group stocks.

Category	Primary grouping	Definition
Release age	Yearling	Fry overwinter in fresh water
	Subyearling	Fry do not overwinter in fresh water
Adult migration timing	Spring	March–May migration
	Summer	June–August migration
	Fall	August–November migration
Juvenile distribution	North	Enter continental shelf north of Vancouver Island
	South	Enter continental shelf south of northern Vancouver Island (excluding Columbia)
	Columbia	Columbia River basin
	Strait of Georgia	Fraser River and east coast Vancouver Island
	Puget Sound	Puget Sound and Juan de Fuca Strait
Adult distribution	Offshore	Rarely caught in pre-terminal fisheries
	North	Typically caught in southeast Alaska and northern BC fisheries
	Broad	Caught along continental shelf from Oregon to southeast Alaska
	Strait of Georgia	Enter Strait of Georgia; typically caught in southern BC and WA fisheries
	Puget Sound	Enter Puget Sound; typically caught in southern BC and WA fisheries

Note: Here, we delineate juvenile and adult distributions at the finest scale considered, but also fit models pooling distributions at coarser scales (described in Table S1).

observation model scaling parameters α are set to zero. Process error is fixed as an identity matrix (i.e., $\mathbf{Q}$ has variance equal to one and zero covariance) and the elements in the first $g - 1$ rows of $\mathbf{Z}$ are set to zero if the column index is greater than the row index to ensure identifiability (Holmes et al. 2014). Finally, we included an autoregressive (AR) component in the process model, rather than assuming trends followed a nonstationary random walk. Thus, eq. 7 becomes

$$(8) \quad \boldsymbol{x}_{t_r} = \boldsymbol{x}_{t_r-1}\boldsymbol{\phi} + \boldsymbol{w}_{t_r} \text{ where } \boldsymbol{w}_{t_r} \sim \text{MVN}(0, \mathbf{Q})$$

where $\boldsymbol{\phi}$ is a vector of length g containing temporal autocorrelation parameters for each trend.

The autoregressive structure is more flexible since it allows the strength of temporal autocorrelation to be directly estimated from the data, rather than fixed. Preliminary analyses also indicated model performance (as estimated by leave-one-out information criteria) was improved for several stock groupings when $\boldsymbol{\phi}$ was estimated. In the case of the Puget Sound subyearling mean age-at-maturity model we used the simpler random walk parameterization due to convergence issues (Fig. S1).

The estimated trends from the DFAs provide a continuous estimate of how shared dynamics have varied through time. When the posterior distribution does not overlap zero, there is a high probability that the trend has diverged from its long-term mean. Similarly, loadings provide an estimate of how closely an individual time series tracks the overall trend. Nevertheless we found that using trends and loadings, which are unitless, to evaluate demographic changes could be difficult, particularly when multiple trends were estimated and observed time series were missing data. Thus we evaluated evidence for nonstationary patterns using additional complementary methods.

First, we fit hidden Markov models (HMMs) to estimate the probability that a given trend g was in one of two mean states (commonly referred to as regimes) over the length of the time series (Zucchini et al. 2017), while explicitly incorporating the uncertainty in the DFA's estimates (Ward et al. 2019). Thus while the estimated trends from the DFA provide evidence for continuous, gradual changes in shared dynamics, the probability states estimated by the HMM provide an indication of when, if ever, the underlying trend transitions to a new mean state. With only two estimated states, this framework is analogous to a breakpoint analysis. While the HMM framework is flexible and a large number of states can be specified a priori, preliminary analyses indicated models with more than two provided qualitatively similar results or failed to converge. We note that although the broader HMM literature refers to alternative states as regimes, they are a function of the time series themselves and do not represent broader environmental regimes per se.

Second, we used the DFAs to generate hindcast predictions of juvenile logit survival rate and mean age-at-maturity. We used these predictions as a qualitative measure of model fit, but also to explore how demographic traits of less data-rich stocks may have varied through time. Predicted time series of logit survival and mean age-at-maturity were centered (i.e., zero corresponds to the observed stock-specific mean), but not scaled, so anomalies are comparable among stocks and stocks groups. We colour-coded stock-specific predictions in our figures based on relative status in recent years (i.e., since 2011 ocean entry year) for each variable. Stocks where the 90% posterior interval of the mean for this period

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overlapped zero are shaded grey. In the online supplement we also present predicted survival in real space and uncentered mean age-at-maturity. Survival estimates in real space are coloured based on the recent predicted mean relative to the predicted mean over the entire time series and its standard deviation. Time series of survival in real space for stocks with uncertain posterior distributions for logit survival remain shaded grey.

Third, we used the hindcast predictions to generate time series of summary statistics–mean estimates within a stock group, as well as the proportion of stocks within a group below their time series mean (defined as greater than 90% probability that the predicted value is different from 0). These summary statistics account for uncertainty in data availability and model estimates to allow inference from the entire suite of stocks, not just those with the longest time series. To reduce interannual variability we calculated group means and stock proportions using a five-year moving window.

We fit dynamic factor models in a Bayesian framework using the bayesdfa package (Ward et al. 2019), which is built upon the Stan statistical language and the No-U-Turn Sampling algorithm (Hoffman and Gelman 2014). To balance interpretability and model performance (in terms of fit to the observed data and therefore the precision of interpolated time series), we estimated two latent trends for each stock grouping identified in the previously described maximum likelihood model selection framework. We initially fit the DFA models using four chains and 4000 iterations, discarding the first 2000 as warmup. We fit the HMM models using one chain (to avoid states being labelled differently among chains) and 3000 iterations, discarding the first 1500 as warmup. If $\widehat{R}$ (a measure of chain convergence) was greater than 1.03 or the effective sample size n_{eff} for any parameter was less than 200, we added 2000 iterations and sampled again. We also assessed that chains were mixing adequately using diagnostic plots.

We recognize that alternative analytical frameworks could be used to address our research questions. For example, a single DFA could be fit to the entire suite of time series, with an optimal number of trends and covariance structure identified via model comparison. However, fitting a large number of Bayesian DFAs to more than 50 time series is computationally intensive. More importantly, the bayesdfa package does not provide a default framework for constraining trends to be assigned to specific time series so groupings must be interpreted post hoc based on patterns of loadings. Thus we feel that our approach balances objective testing of a priori hypotheses with Bayesian inference applied to the final model, which allows for flexible sampling from posterior distributions.

Finally, we completed a supplementary analysis to address two related questions. First, given that both juvenile survival rate and mean age-at-maturity are model derived quantities that depend on coded wire tag recoveries, are they independent time series or do they strongly covary with one another? Second to what extent are patterns in juvenile survival rate and mean age-at-maturity driven by hatchery release size?

Results

The most parsimonious models for both juvenile survival rate and mean age-at-maturity grouped stocks by juvenile marine distribution (Table 2, Tables S2 and S3). The best supported juvenile distribution model for both response variables was relatively fine scale, accounting for juvenile outmigration age (yearling vs. subyearling) and distinguishing between northern, Strait of Georgia, Puget Sound, and southern (i.e., westcoast Vancouver Island, coastal Washington/Oregon, Columbia River) stock groupings. More complex models that split Columbia River populations from other southern stocks that enter the California Current were not supported.

We next used dynamic factor analysis in a Bayesian framework to estimate two trends for each response variable and juvenile distribution grouping. We excluded three groupings, northern subyearling, Strait of Georgia yearling, and southern yearling, which had too few stocks to fit a DFA, resulting in five juvenile survival rate models and five mean age-at-maturity models, each with two estimated latent trends.

Some, but not all, stock groupings showed evidence of reduced survival rates relative to the late 1970s (Figs. 2, 3, S2). The first trend in logit juvenile survival rate for Strait of Georgia subyearling stocks declined early in the time series, reaching a minimum in the early 1990s, before increasing slightly (Fig. 2). There was a high probability of a single transition (from high to low survival mean state) for the first Strait of Georgia subyearling trend in the mid-1980s. The first trends for southern subyearling and northern yearling stocks exhibited relatively more variability. Here, juvenile survival rates cycled between below- and above-average levels. Conversely, the first trend for both Puget Sound yearling and subyearling stocks was stationary, providing no evidence that juvenile survival rates were previously uniformly high in this region and no evidence of a change in the mean state. Among stock groups with evidence of a change in mean state, differences in survival between the high and low periods ranged from 1% to 8% (Fig. 4). The second trends in logit juvenile survival rate for northern and Puget Sound yearling, as well as Strait of Georgia subyearling, stock groupings were generally more stable through time (Fig. 2). Puget Sound and southern subyearlings showed relatively more variability; however, none of the second trends in juvenile survival rate showed evidence of a state switch.

Estimated loadings (i.e., the extent to which a given stock's dynamics were represented by a shared trend) were typically greater for the first trend; however, there was substantial variation in strength and direction of loadings within a given stock grouping (Fig. S3), consistent with variation among stocks in estimated fits (Fig. 3). Posterior predictions from each stock group's DFA indicated mean logit survival was poorest in the mid-1990s for all subyearling groups (Fig. 4). Minima for yearling groups occurred in recent years, but were more uncertain (Fig. 4).

Trends in mean age-at-maturity also varied among stock groupings, ranging from marked declines to relative stability. At least one, and often both, trends declined among all subyearling stocks, though the timing of these changes varied (Fig. 5). The first trend for Strait of Georgia subyearlings

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Table 2. Relative performance of the five most parsimonious maximum likelihood state-space models.

Response	Ecological grouping	ΔAIC_c	AIC_c weight
Survival rate	Juvenile Distribution Fine (N/SoG/PS/S) + Release Age	0.00	0.84
	Juvenile Distribution Very Fine (N/SoG/PS/WC/CR) + Release Age	6.25	0.04
	Juvenile Distribution Very Fine (N/SoG/PS/WC/CR)	6.31	0.04
	Juvenile Distribution Fine (N/SoG/PS/S)	6.63	0.03
	Juvenile Distribution Coarse (N/SS/S) + Release Age	21.47	<0.01
Mean age-at-maturity	Juvenile Distribution Fine (N/SoG/PS/S) + Release Age	0.00	0.73
	Juvenile Distribution Very Fine (N/SoG/PS/WC/CR)	3.14	0.15
	Juvenile Distribution Very Fine (N/SoG/PS/WC/CR) + Release Age	6.19	0.03
	Juvenile Distribution Very Coarse (CS/SS)	6.22	0.03
	Juvenile Distribution Fine (N/SoG/PS/S)	6.27	0.03

Note: Each model grouped latent processes in juvenile survival rate or mean age-at-maturity based on stock traits. Juvenile and adult distribution groupings ranged from coarse- to fine-scaled, with regional groupings abbreviated in parentheses. Results for full model set are shown in Tables S2 and S3. Regional abbreviations associated with distribution groupings: south (S), continental shelf (CS), Columbia River (CR), north (N), Puget Sound (PS), Salish Sea (SS), Strait of Georgia (SoG), OR/WA/WCVI coastal (WC).

Fig. 2. Estimated trend one (A) and trend two (C) for logit juvenile survival rate models and model predicted median probability of each trend (B, D) being in a high survival mean state, relative to its long-term mean. Ribbons represent 90% posterior intervals. Colours represent age at release. Stocks are arranged by groupings (approximately north (left) to south (right)). [Colour online.]

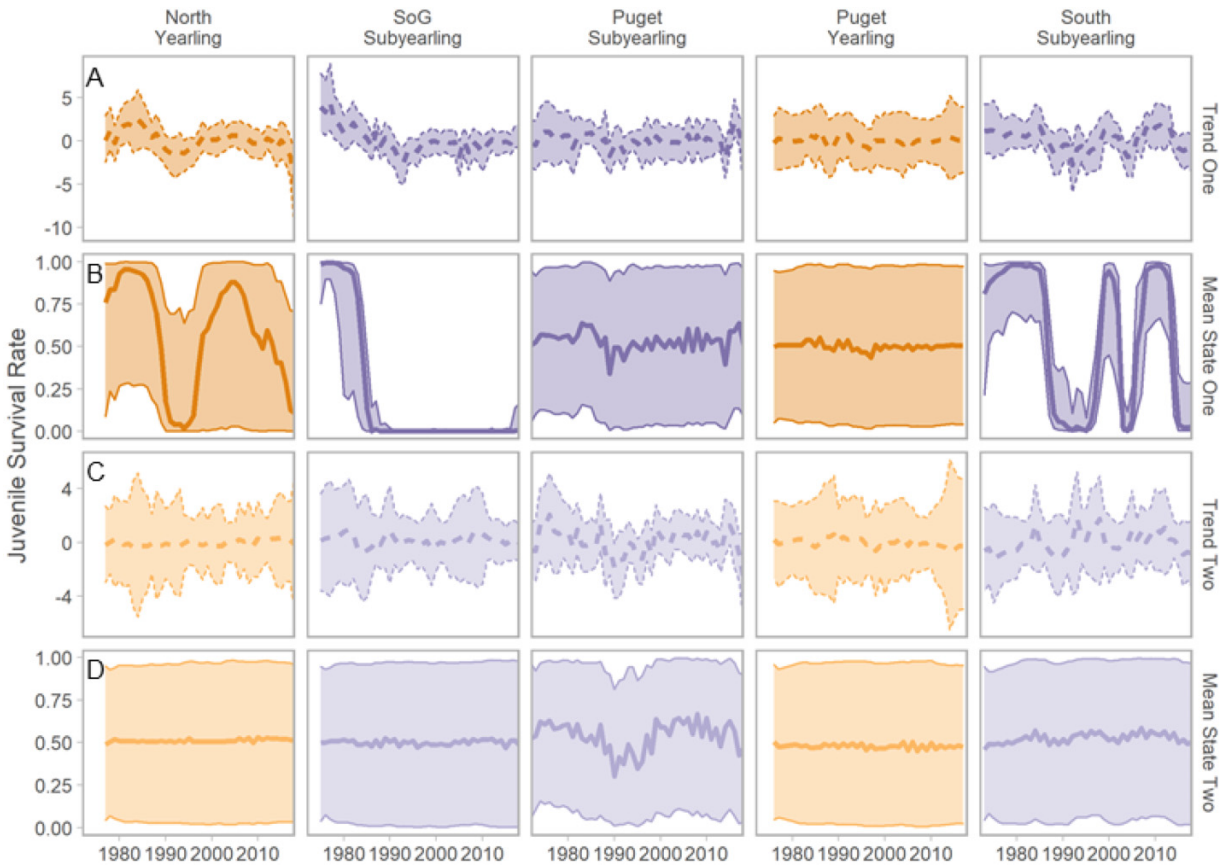
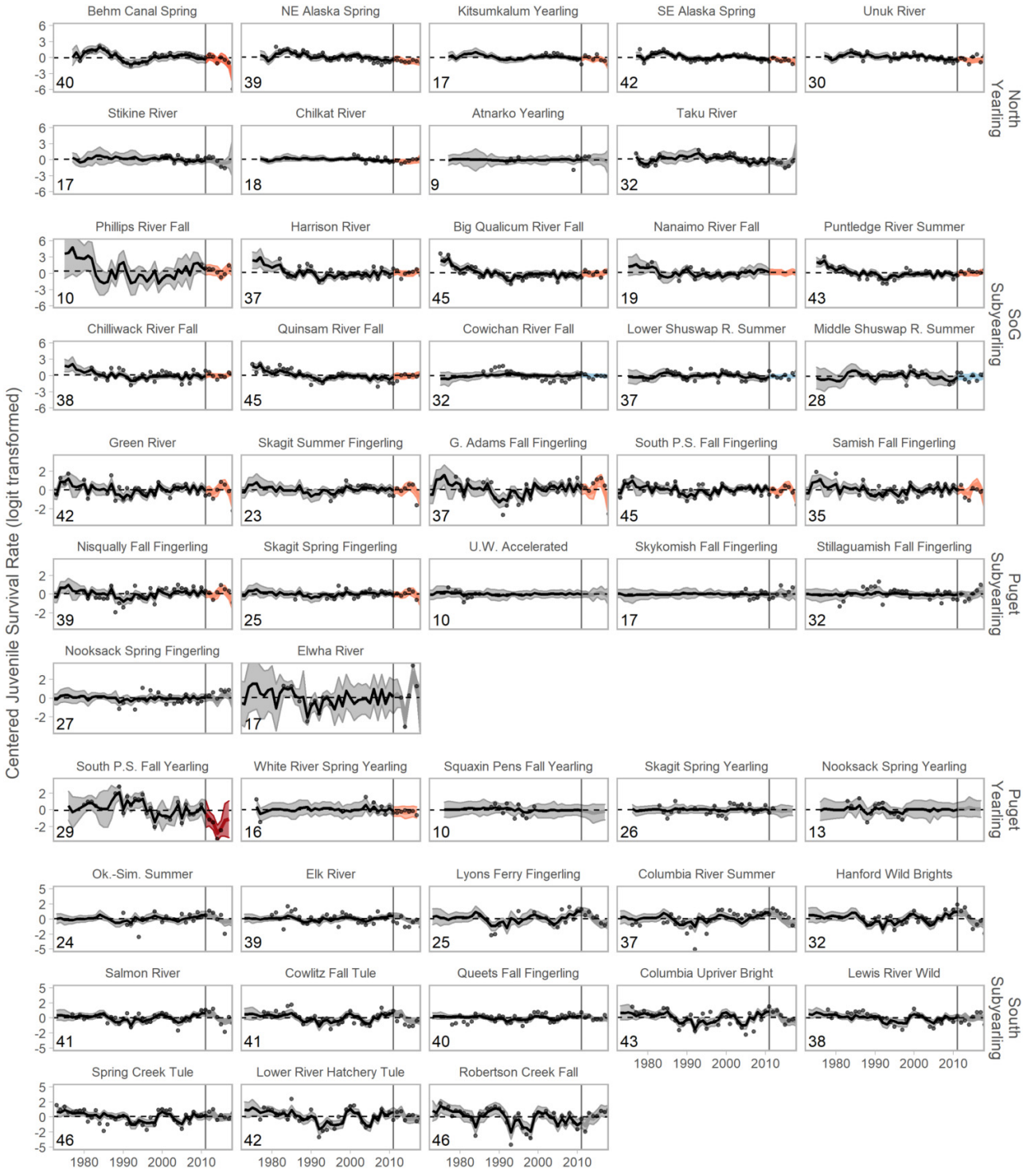


Fig. 3. Model predicted mean (lines) and observed (points) stock-specific juvenile survival rates (logit-transformed and centered). Ribbons represent 90% posterior intervals. Horizontal dashed lines represent the mean predicted stock-specific logit juvenile survival rate. Lines and ribbons since 2011 (vertical line) are coloured so that warm and cool colours represent negative and positive, respectively, mean logit survival rate anomalies during this period (stocks where the 90% posterior interval of the post-2011 mean overlapped zero are shown in grey). Stocks are arranged by groupings (approximately north to south) and by ascending recent mean logit survival rate. *Y*-axis range and predicted time series length differ among, but not within, stock groupings. The number of annual observations for each stock is in the lower left. [Colour online.]



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Fig. 4. Model predicted among stock mean juvenile survival rate. Points represent the rolling five-year mean based on posterior predictions (uncentered and back-transformed) from the dynamic factor model and whiskers represent 90% intervals. Blue (red) points represent five year periods where the among stock mean centered logit survival rate was greater (less) than the long-term mean (i.e., zero). When the estimated mean overlapped zero, points are grey. [Colour online.]

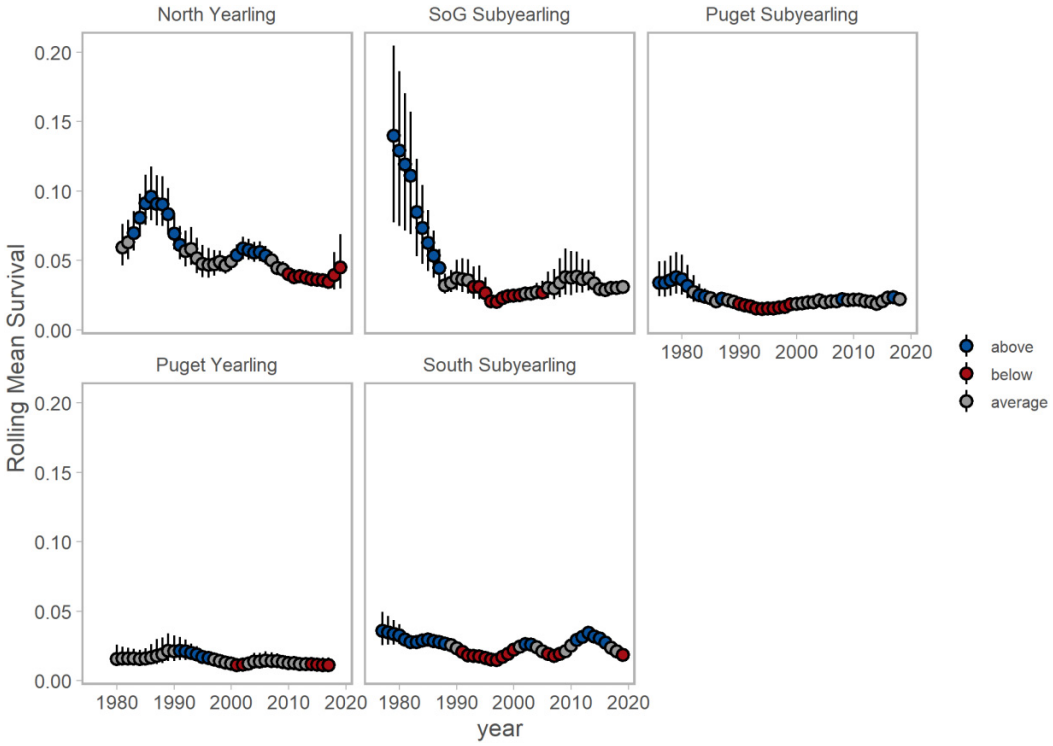
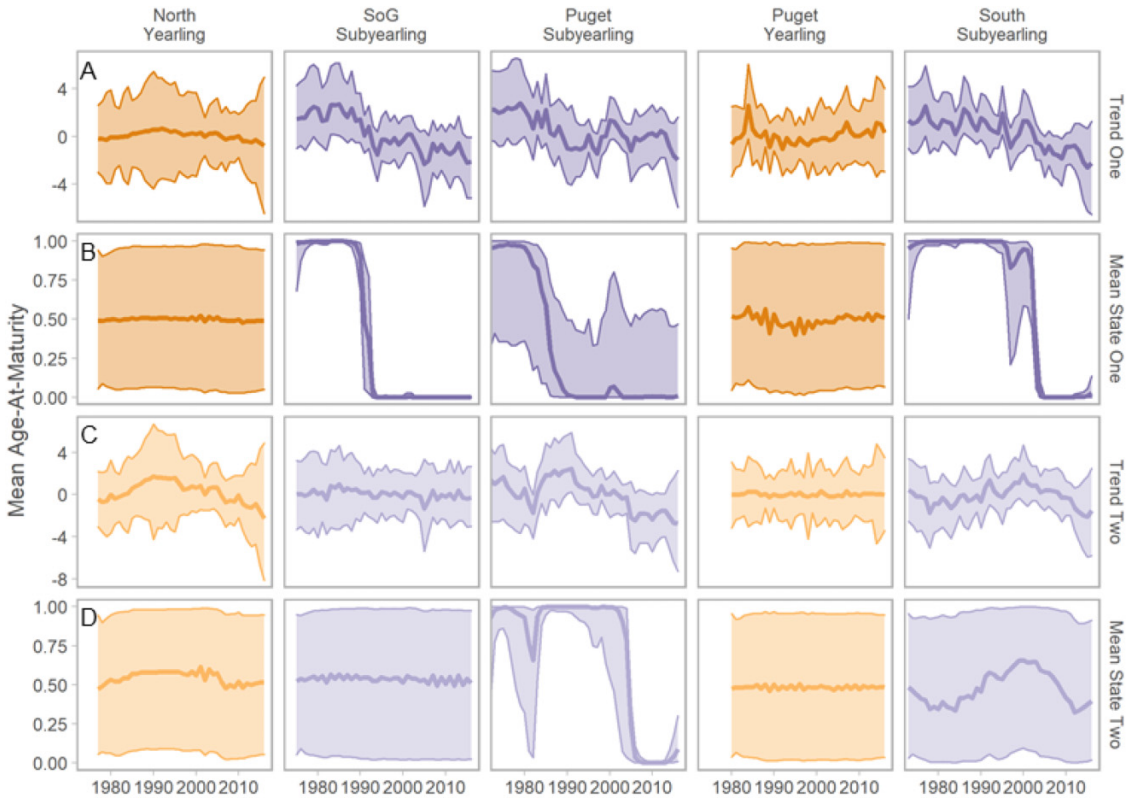


Fig. 5. Estimated trend one (A) and trend two (C) for mean age-at-maturity models and model predicted median probability of each trend (B, D) being in an older mean state, relative to its long-term mean. Ribbons represent 90% posterior intervals. Colours represent age at release. Stocks are arranged by groupings (approximately north (left) to south (right)). [Colour online.]



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showed evidence of a marked reduction in mean age-at-maturity beginning in the 1990s and southern subyearling age-at-maturity declined approximately a decade later. Both Puget Sound subyearling trends showed evidence of a decline, with the first trend declining in the mid-1980s and the second in the mid-2000s (Fig. 5). Each subyearling grouping showed evidence of a shift from an older to a younger age-at-maturity mean state. Model predictions indicated mean age-at-maturity within each subyearling group was 0.23–0.31 years younger after these shifts (Fig. 7) and declines in individual stocks were often more severe (Fig. 6, Fig. S4).

Conversely, the estimated trends for northern yearling stocks showed some evidence of a decline in mean age-at-maturity, but these changes were relatively modest and not associated with a change in mean state (Fig. 5). Puget Sound yearling trends were stationary. We note that despite these trends, several yearling stocks showed evidence of marked declines in mean age-at-maturity (Fig. 6, Fig. S4) and the overall mean for both yearling groups is below its peak in the early (northern yearling) or late (Puget Sound yearling) 1990s (Fig. 7). Estimated loadings for trends in mean age-at-maturity were more homogeneous than juvenile survival rate loadings, suggesting greater synchrony within each stock group (Fig. S5).

Recent and significant negative anomalies in Chinook salmon age-at-maturity were relatively more common than declines in juvenile survival. Between 2011 and 2016 (the last ocean entry year for which complete age composition data were widely available), 23% to 77% of the stocks within each group had above average logit survival in recent years (Fig. 8). With the exception of Puget Sound yearling stocks, less than 20% of the stocks within each group had above average mean age-at-maturity in recent years (Fig. 8).

A supplementary analysis indicated trends in juvenile survival and mean age-at-maturity persisted after accounting for correlations between survival and age (online Supplement). Similarly temporal trends also persisted after accounting for the effect of variability in size at release (online Supplement).

Discussion

Declines in Chinook salmon abundance have resulted in negative impacts to First Nations, recreational, and commercial fisheries (Lindley et al. 2009; ADF&G Chinook Salmon Research Team 2013; Doutaz et al. 2021). Indeed there is growing concern that some populations may not persist without significant interventions that extend beyond managing harvest rates (NMFS 2016; Crozier et al. 2021; COSEWIC 2018). While the weeks following ocean entry are frequently considered a critical period to Pacific salmon recruitment (Ruff et al. 2017; Beamish and Mahnken 2001), poor Chinook salmon productivity may also be a consequence of declines in body size and age-at-maturity (Oke et al. 2020; Ohlberger et al. 2018; Manishin et al. 2021).

We built upon previous analyses of Chinook salmon population dynamics to identify ecological groupings that are associated with shared temporal patterns in juvenile survival rate and mean age-at-maturity (a proxy for both maturation age and late stage mortality), and to compare how these two characteristics vary through time and among stocks. Both juvenile survival rate and mean age-at-maturity covaried most strongly with juvenile marine distribution, which divided 57 Chinook salmon stocks into eight stock groupings based on ocean entry location and release age (yearling or subyearling). Juvenile survival rates are almost universally low; however, many stocks in the Strait of Georgia and Puget Sound do not show evidence of a recent decline, exhibiting stationary survival rates since the early 1980s. Conversely mean age-at-maturity appeared to still be trending downward for many stock groups. All subyearling stock groupings showed evidence of a shift in mean state consistent with younger age-at-maturity, but asynchronously, and a greater proportion of indicator stocks have exhibited significantly younger ages in the most recent years than below average survival. Below we discuss why both survival and age-at-maturity may covary with juvenile marine distributions, describe differences among regions in demographic changes, and explore the management implications of divergent patterns in these two demographic traits.

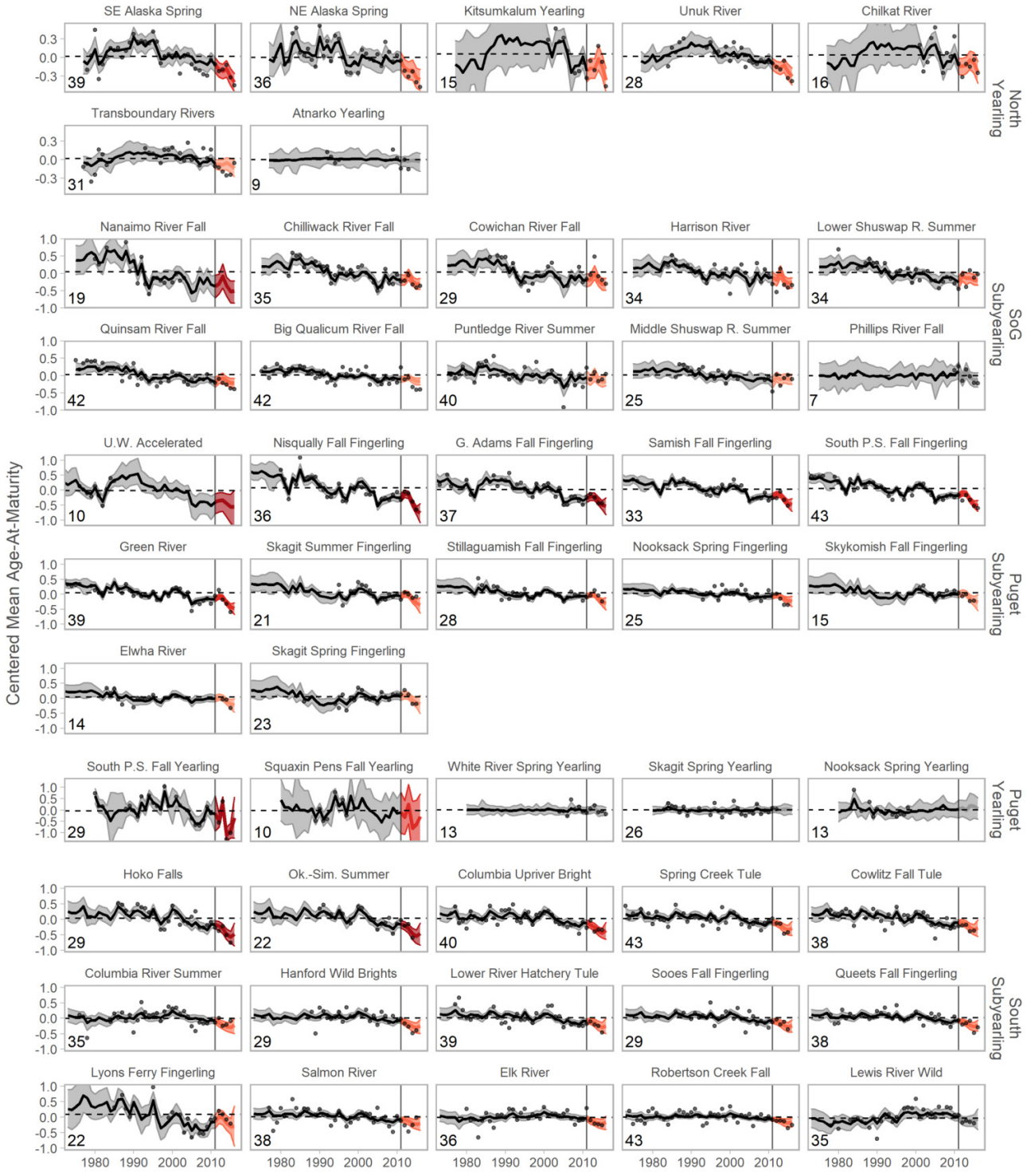
Ecological traits associated with stock groupings

We evaluated support for state-space models that modeled variation in juvenile survival rate or mean age-at-maturity as an underlying process shared among stocks with a common (*i*) release age (a proxy for freshwater life history in wild populations), (*ii*) run timing group, (*iii*) juvenile marine distribution, or (*iv*) adult marine distribution. For the distribution models we considered multiple scales of complexity, ranging from coarse to relatively fine spatial clustering, as well as an interaction with release age. For both response variables, the most parsimonious model grouped stocks based on juvenile distribution at an intermediate scale—separating stocks depending on ocean entry location into northern (north of Vancouver Island), Strait of Georgia, Puget Sound, and southern (west coast Vancouver Island, coastal Washington/Oregon, and Columbia River) groupings—while also accounting for release age.

Linkages between ocean entry location (a proxy for juvenile marine distribution) and juvenile survival rates are intuitive given that many Chinook salmon populations disperse along the continental shelf relatively slowly (Tucker et al. 2011; Fisher et al. 2007) or remain resident in near-shore areas (Chamberlin et al. 2011; Shelton et al. 2019; Freshwater et al. 2021). That trends in age-at-maturity also mapped to juvenile distributions was more surprising. Age-at-maturity is typically associated with late-stage growth and mortality (Quinn 2018). Thus we anticipated age-at-maturity would covary most strongly with adult marine distributions.

It is possible that environmental conditions during the first year of ocean residence have persistent carry-over effects that impact later life stages (Xu et al. 2020). Alternatively, juvenile and adult groupings overlap in many cases, and these relatively coarse classifications may obscure subtle stock-specific differences in distribution during life stages where age-at-maturity is determined (Shelton et al. 2019; Freshwater et al. 2021). A third possibility relates to late-stage mortality.

Fig. 6. Model predicted mean (lines) and observed (points) stock-specific centered mean age-at-maturity. Ribbons represent 90% posterior intervals. Horizontal dashed lines represent the mean predicted mean age-at-maturity. Lines and ribbons since 2011 (vertical line) are coloured so that warm and cool colours represent negative and positive, respectively, mean age-at-maturity anomalies during this period (stocks where the 90% posterior interval of the post-2011 mean overlapped zero are shown in grey). Stocks are arranged by groupings (approximately north to south) and by ascending centered mean age-at-maturity since 2011. Note that y-axis scale and predicted time series length differ among, but not within, stock groupings. The number of annual observations for each stock is in the lower left. [Colour online.]



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Fig. 7. Model predicted among stock mean of mean age-at-maturity. Points represent the rolling five-year mean based on posterior predictions (uncentered) from the dynamic factor model and whiskers represent 90% intervals. Blue (red) points represent five year periods where among stock mean age-at-maturity was greater (less) than the time series mean. When the 90% interval of the mean overlapped the time series mean, points are grey. [Colour online.]

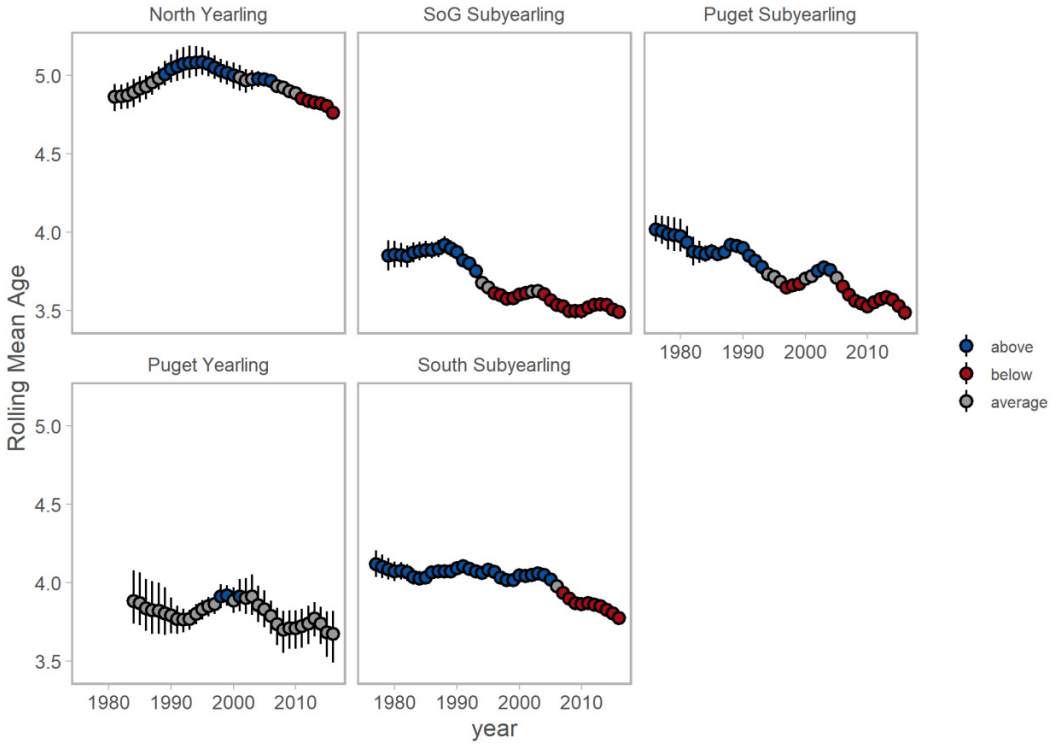
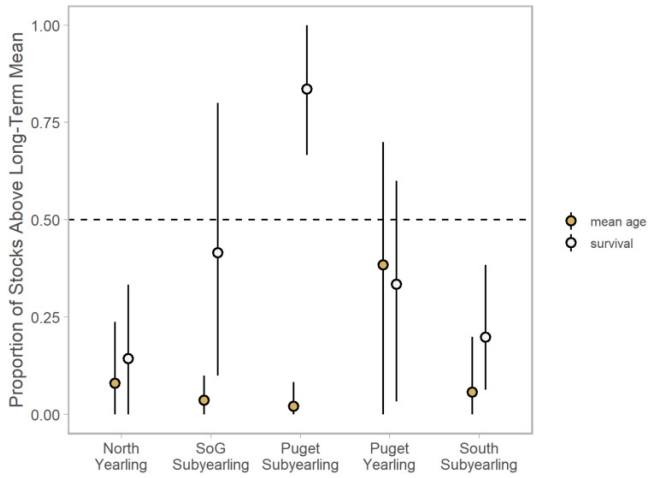


Fig. 8. The proportion of stocks within a grouping with estimated logit survival or mean age-at-maturity above the stock-specific time series average. Points represent the mean proportion from 2011-2016 (last five year period with complete data for both datasets) based on posterior predictions from the dynamic factor model. Whiskers represent 90% intervals.



Because we defined juvenile distributions based on ocean entry location, these clasifications also reflect differences among stocks in terminal portions of adult marine migrations. If impacts on age-at-maturity occur late in life when populations are returning to spawn, for example due to predation when fish are concentrated in near-shore areas, then correlations with ocean entry location might be expected, but independent of juvenile ecology. Finally, there is growing evidence that age-at-maturity is underpinned by genetic variation and juvenile distribution may simply be the best proxy among our covariates for relatedness (McKinney et al. 2020, 2021).

Regionally coherent trends in survival and age-at-maturity are consistent with evidence that salmon population dynamics covary at intermediate spatial scales (Dorner et al. 2017; Mueter et al. 2002; Ruff et al. 2017). Nevertheless we identified several caveats to this general pattern. First, as previously highlighted by Ruff et al. (2017), we found support for distinct trends for Strait of Georgia and Puget Sound stocks despite their close geographic proximity. Both stock groups encounter considerable anthropogenic impacts and high predator densities (Nelson et al. 2018; Sobocinski et al. 2021); however, only Strait of Georgia stocks showed evidence of a decline in survival since the 1980s. The timing of declines in age-at-maturity for these two stock groups also differed. Distinct trends in survival and maturation age may indicate

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localized processes shape dynamics, for example due to fine-scale differences in spatial distribution (Arostegui et al. 2017; Freshwater et al. 2021; Quinn and Losee 2022).

Second, Columbia River stocks did not exhibit evidence of divergent dynamics relative to other southern populations (westcoast Vancouver Island, coastal Washington, and coastal Oregon populations) that directly enter the California Current. There is evidence of substantial carryover effects on mortality associated with downstream migration for some Columbia River stocks (Gosselin et al. 2018; Chasco et al. 2021); however, our analysis included relatively few Upper Columbia River and Snake River populations that are most strongly impacted by these effects. Lower Columbia River populations may naturally covary with fall run, subyearling coastal populations.

Third, we found that subyearling and yearling stocks originating from the same geographic region exhibited distinct trends in age-at-maturity and juvenile survival. Yearling Chinook salmon from the Upper Columbia River basin migrate rapidly after ocean entry compared to other stocks (Fisher et al. 2014). Yet other Columbia River stocks, as well as stocks from coastal areas and the Salish Sea, have broadly overlapping marine distributions regardless of freshwater life history (Trudel et al. 2009; Tucker et al. 2011; Beamish et al. 2011). Unique trends among yearling and subyearling stocks may reflect fine-scale differences in habitat use, such as ocean entry phenology (Bourret et al. 2016; Quinn and Losee 2022) or the timing of offshore or northward migrations during their second marine year (Tucker et al. 2011). Mechanisms resulting in divergent survival may also include size-selective vulnerability to predators (Nelson et al. 2021) and differential exposure to pathogens (Tucker et al. 2018).

Regional variation in juvenile survival rate

From southeastern Alaska to central California, Chinook salmon marine survival has been low, on average, for several decades (Welch et al. 2021). However, we found that evidence for nonstationary survival varies strongly among stock groups, emphasizing that periods of low survival cannot necessarily be conflated with declining survival. For example, neither of the estimated survival trends for southern subyearling stocks was consistent with a transition from high to low survival. Instead the first trend exhibited cyclical variability, oscillating between above- and below-average survival periods and the second trend was effectively stationary. Similarly, both Puget Sound yearlings and subyearlings have had long-standing (40–50 years) periods of low survival, which again highlights the need to differentiate between low survival and declining survival, especially when considering management actions intended to rebuild stocks. Conversely, we identified more robust declines in the first survival trend for Strait of Georgia subyearling and northern yearling stock groups. The former showed evidence of a shift from a high to low survival mean state, with minima occurring during the early 1990s. Northern yearling stocks showed a more nuanced pattern, with survival improving initially, then declining in the 1990s. The decline accelerated in recent years, but there is still equivocal evidence for a change in the mean survival state.

Patterns in Chinook salmon juvenile survival rates through time likely reflect differences in how recruitment dynamics are regulated among regions. The cyclic juvenile survival rate observed among populations that enter the California Current is consistent with evidence linking changes in the productivity of these stocks to low-frequency variation in basin-scale processes such as the Pacific Decadal Oscillation and North Pacific Gyre Oscillation (Kilduff et al. 2014; Hertz et al. 2016; Mantua et al. 1997). Atmospheric forcing may mediate Chinook salmon survival via changes in prey abundance or quality, as well as growth rate (Miller et al. 2014; Burke et al. 2013). Relatively robust relationships with physical mechanisms that impact trophic interactions—such as linkages between upwelling, salmon prey, and salmon predators—have also been identified for several California Current stocks (Wells et al. 2016). Conversely, identifying environmental conditions driving variation in survival rates has proven more difficult for stocks with evidence of a single transition between high and low mean states, or when survival is low, but stable. The marine survival of Puget Sound stocks appear to be associated with a diverse suite of environmental processes; however because survival rates have been persistently low, there is relatively little contrast in the data and the predictive power of mechanistic models is modest (subyearling stocks) or negligible (yearling stocks) (Sobocinski et al. 2021). Studies attempting to identify drivers of juvenile survival in the Strait of Georgia during the low survival period have been similarly constrained (Perry et al. 2021). Harbour seal predation has been identified as a factor influencing Chinook salmon survival in the Salish Sea (Nelson et al. 2018). Nevertheless, it is unclear why some Strait of Georgia subyearling stocks have recently exhibited record high escapements (CTC 2021*b*) or why seal predation would create a shift in survival rates in the Strait of Georgia, but not Puget Sound, since seal abundance initially increased in each region at similar rates (Nelson et al. 2018). Substantial uncertainty remains about which salmon life stages are most vulnerable to seal predation, and the population-level impacts of this predation, given evidence of size-selective predation (Nelson et al. 2021). Identifying mechanistic drivers of changes in the survival of Salish Sea populations may be intrinsically more difficult because both Puget Sound and Strait of Georgia populations may rear and mature in near-shore waters (Chamberlin et al. 2011; Freshwater et al. 2021). Since the proportion of resident fish appears to vary interannually (Quinn and Losee 2022), linkages between drivers and survival may be obscured by differential migration.

Regional variation in mean age-at-maturity

Although declines in the mean age-at-maturity of Chinook salmon stocks are widespread (Ohlberger et al. 2018), they are not uniform. Instead the timing and extent of declines vary among regions and life history types. In the case of northern yearling stocks, declines in age have not stabilized in recent years; however, their absolute magnitude is fairly modest (less than four months relative to the long-term mean). Subyearling stocks in the Strait of Georgia, Puget Sound,

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and California Current regions show relatively larger declines with mean ages-at-maturity that are six months to a year younger than the long-term mean. However the timing of these declines differs among the groupings, ranging from the mid-1980s to the mid-2000s. Trends for Puget Sound yearling stocks were stationary, suggesting weaker evidence for synchronous changes in age-at-maturity.

Our observations also indicate declines in age-at-maturity among British Columbia populations may be larger than previously reported (Ohlberger et al. 2018; Oke et al. 2020). Oke et al. (2020) found minimal evidence for changes in size and age of southeast Alaska Chinook salmon and suggested this might be due to data originating from troll fisheries in that region, which are dominated by immature fish from outside Alaska with weaker evidence of declines. Conversely, our results indicate declines in mean age-at-maturity of BC, Washington, and Oregon populations are widespread and often stronger than those of southeast Alaskan populations. This discrepancy may be due to previous authors using estimates of age-specific abundance directly from CWT recoveries while our analysis accounted for variability in age-specific harvest rates by using estimates from the cohort analysis.

Earlier maturation of Pacific salmon may be caused by a range of processes. For example, declines in Alaskan Chinook salmon and Washington chum salmon age-at-maturity were correlated with interspecific competition at a basin-scale (Oke et al. 2020; Litz et al. 2021). However in this study, subyearling stocks exhibited the strongest declines in age-at-maturity. Most subyearling Chinook salmon populations in southern regions spend their marine lives on the continental shelf, overlapping with pink, chum, and sockeye salmon (*Oncorhynchus nerka*) only briefly after ocean entry (Weitkamp 2010; Freshwater et al. 2021). Thus, interspecific interactions seem less plausible as a coastwide driver of declines in age-at-maturity in Chinook salmon. Indeed, Oke et al. (2020) note that evidence for declines in Chinook salmon body size persisted after accounting for density dependent effects.

Alternatively, favorable environmental conditions and increased growth may be correlated with earlier maturation in salmonids (Quinn 2018). Declines in mean age-at-maturity for several western Alaskan Chinook salmon populations were correlated with increases in growth rates during the second ocean year, which were in turn associated with warmer sea surface temperatures (Siegel et al. 2017). Reaction norms in these stocks also changed so that relatively less marine growth was required to reach a threshold at which maturation occurred (Siegel et al. 2018). Nevertheless declines in maturation age have also been observed in Salish Sea and California Current stocks that do not show a concomitant increase in marine growth (Duffy and Beauchamp 2011; Daly and Brodeur 2015; Xu et al. 2020; Claiborne et al. 2021).

Juvenile size at ocean entry is correlated with age-at-maturity in both hatchery and wild Chinook salmon populations, suggesting age-at-maturity can also be moderated by freshwater conditions via carryover effects (Scheuerell 2005; Spangenberg et al. 2015; Tattam et al. 2015). Hatchery practices can explicitly amplify these processes, such as when warmer groundwater is used for fish rearing, resulting in greater overwinter growth and higher jacking rates (Harstad et al. 2018). Finally, because maturation age is heritable and associated with discrete genetic architecture (McKinney et al. 2021), hatchery practices may result in selective pressures that reduce underlying age class diversity. Most indicator stocks are hatchery-origin, and it is possible that declines in age have been amplified by these, or other, processes, but this assumption should be weighed carefully given evidence of declines in regions where hatchery impacts are minimal (e.g., Alaska (Oke et al. 2020), the Harrison River, BC (Doutaz et al. 2021)). Although limited in scope, our supplementary analysis also indicated temporal trends in age-at-maturity persisted after accounting for variability in hatchery release size.

On the other hand, changes in natural mortality could cause an actual decline in age-at-maturity due to differences in selective pressures, as well as apparent declines due to necessary assumptions in the cohort reconstruction. Several lines of evidence suggest natural mortality rates late in life may be higher than previously assumed. First, tagging studies have identified considerable mortality of age-3+ Chinook salmon in coastal regions (Seitz et al. 2019; Courtney et al. 2019) and during migrations through estuaries (Wargo Rub et al. 2019). In both cases mortality appears to be associated with predation (e.g., sleeper sharks (*Somniosus pacificus*) and salmon sharks (*Lamna ditropis*) in the Gulf of Alaska; sea lions (*Zalophus californianus* and *Eumatopias jubatus*) in the Columbia River estuary). Second, simulation models indicate increased natural mortality of older Chinook salmon age classes is a realistic driver of observed changes in age structure over time (Manishin et al. 2021). Third, both bioenergetics and population models suggest increases in the abundance of Chinook salmon predator populations (e.g., northern resident killer whales, *Orcinus orca*) could lead to substantial increases in salmon mortality rates at older ages, more than offsetting reductions in harvest (Chasco et al. 2017*b*; Ohlberger et al. 2019).

If predation is the primary driver of changes in Chinook salmon age-at-maturity then disparate trends among regions and life history types suggest stocks interact with distinct predator communities or differ in their exposure to common predators. Resident killer whales are selective predators that may disproportionately impact Chinook salmon (Ward et al. 2009; Ford et al. 2010; Chasco et al. 2017*b*). Both resident killer whales and Chinook salmon are highly mobile and contain populations with distinct distributions, which makes estimating predation risk difficult. Nevertheless declines in age-at-maturity among subyearling stocks suggest they may experience particularly high predation risk. The marine distributions of many subyearling stocks are centered on the west coast of Vancouver Island (Shelton et al. 2019), which may expose them to predation by northern resident killer whales as they mature (Riera et al. 2019), as well as southern resident killer whales during summer months prior to spawning (Hanson et al. 2021, 2010). Other piscivores such as salmon sharks may also be a significant source of mortality (Seitz et al. 2019; Manishin et al. 2021; Okey et al. 2007), but estimates of predation rates are not possible due to limited data on abundance, distribution, and diet.

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Critically, Chinook salmon age-at-maturity in this context may be driven by either earlier maturation or increased natural mortality after fish are vulnerable to harvest (second or third year of ocean residence depending on age at release). This is because the CTC cohort analysis which generates these data assumes that natural mortality after recruitment is fixed and cannot distinguish between the two processes (CTC 2021*a*). Given the wide range of mechanisms, in both freshwater and marine habitats, that may drive changes in mean age-at-maturity, structuring future work to determine the relative importance of earlier maturation versus increased mortality will provide critical information as to which drivers are most plausible.

Limitations

A key assumption of this and many other analyses of Chinook salmon dynamics is that the dynamics of hatchery-origin indicator stocks reflect those of wild populations, which are generally the target of conservation policy, but are logistically more difficult to tag in large numbers. Some evidence indicates that covariability in the productivity of wild populations has less spatial structure than that of hatchery stocks (Dorner et al. 2017). The marine survival of wild individuals may differ from those reared in hatcheries (Melnychuk et al. 2014) or appear to differ due to differences in marking methodology (Irvine 2021). Indeed, the abundance of some Chinook salmon populations has increased as the proportion of wild spawners has risen (e.g., Cowichan River; CTC 2021*b*). On the other hand, wild populations with long time series generally have shown comparable declines in smolt-to-adult survival as hatchery populations (Welch et al. 2021). Furthermore, declines in Chinook salmon size and age appear to be widespread and include regions with minimal hatchery influence (Ohlberger et al. 2018; Oke et al. 2020; Xu et al. 2020). While these similarities suggest our findings may be broadly representative, we encourage investigations into the robustness of hatchery indicator stocks as valid proxies of wild Pacific salmon population dynamics. We also note that our analysis was constrained to the central portion of Chinook salmon's North American range, excluding northern and southern populations. Although previous work suggests western Alaskan and California Chinook salmon populations have also declined in abundance and size-at-age (Ohlberger et al. 2016, 2018; Munsch et al. 2022), our findings may be less relevant to these regions.

There are also concerns that the precision of estimates based on CWT data has declined through time due to mark-selective fisheries, reduced exploitation rates, and other factors (Hankin et al. 2005; Moran et al. 2018). These changes may increase the uncertainty of exploitation rate estimates, and therefore survival and age estimates, but appear unlikely to bias estimates of survival or age-at-maturity towards lower values. A detailed evaluation of the impacts of variations in the quantity and quality of CWT data was beyond the scope of this analysis (but see Pacific Salmon Commission Joint CWT Implementation Team 2015).

Finally, shared trends can provide a unifying framework for understanding general changes in demographic traits among a large number of Chinook salmon stocks across a broad geographic area. Nevertheless there is substantial variability among stocks within a given grouping that we cannot account for. For example, the Chilliwack stock appears to have survival rates nearly an order of magnitude higher than other Strait of Georgia subyearling stocks. The South Puget Sound Fall Yearling stock is another outlier, showing steep declines in age-at-maturity that are not shared by other yearling Puget Sound populations. In general the effects of common environmental drivers on Chinook salmon dynamics can vary even at relatively small spatial scales (e.g., Nelson et al. 2018; Xu et al. 2020). Future research should investigate whether divergent patterns are driven by granular differences in marine distribution, significant carry-over effects of freshwater residence, hatchery rearing practices, or other factors.

Management implications

Pacific salmon researchers, managers, and stakeholders have identified reducing, or offsetting, contemporary levels of juvenile marine mortality as key to rebuilding Chinook salmon populations (Pearsall et al. 2021; Welch et al. 2021). However; our results emphasize that the high survival period of the 1970s and 1980s may have been constrained to a relatively small number of stocks, while the majority have exhibited stable or cyclical survival for decades despite changing environmental conditions (e.g., Amaya et al. 2020; Litzow et al. 2020).

Over the long term, the declines in mean age-at-maturity we observed will likely reduce the strength of portfolio effects associated with age class diversity, increasing interannual variability in Chinook salmon abundance (Schindler et al. 2010; Munsch et al. 2022). More immediate impacts include reductions in fecundity, egg size, reproductive success, and, ultimately, the speed at which Chinook salmon populations rebuild, even if marine survival can be improved (though reduced generation time may partially offset these negative effects). Failure to consider the effects of declines in size and age-at-maturity may result in management benchmarks that are insufficiently conservative (Ohlberger et al. 2020; Marshall et al. 2021; Staton et al. 2021). For example, if the number of recruits produced per spawner is positively correlated with spawner age or size, then a given escapement target will be less productive as mean age or size declines. The extent to which changes in age-at-maturity may impact the annual assessment process used by the CTC to identify allowable catches is difficult to estimate. On the one hand, maturation rates are explicitly incorporated into the assessment model so that allowable catches reflect the estimated abundance of different age classes (Peterman et al. 2016). However, the assessment model also depends on forecasts of abundance that are generated by the individual agencies responsible for managing specific indicator stocks. Forecast models differ among agencies and stocks, and the extent to which they are robust to changes in mean age-at-maturity also varies (Peterman et al. 2016).

Predicting the population-level effect of changes in age-at-maturity is not straightforward because fecundity is strongly correlated with size and varies widely among populations

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(Healey and Heard 1984). Yet even after accounting for body size, a one year difference in maturation age was associated with 30% decrease in egg size for University of Washington hatchery Chinook salmon (Quinn et al. 2004). Thus declines in age structure and body size may act synergistically to depress reproductive potential. Importantly, the magnitude and reversibility of declines in mean age-at-maturity will depend on the extent to which they are driven by increased late-stage natural mortality or younger maturation.

Population productivity is driven by interactions among multiple demographic traits. Both poor survival and declining age-at-maturity will limit population productivity, but the relative impact of each process on rebuilding timelines for exploited species is rarely quantified. This shortcoming fundamentally limits our ability to identify optimal management strategies, particularly when conservation resources are finite and the need for interventions appear pressing. One clearly desirable management goal essential to the long-term viability of Pacific salmon, especially in the context of uncertain future environmental conditions, is maximizing life history diversity by maintaining large numbers of evolutionary distinct stock groups (DFO 2005; Freshwater et al. 2019; Sullaway et al. 2021).

We believe the best path forward involves two parallel processes. The first is to develop novel data streams to estimate key demographic parameters for Chinook salmon that are currently poorly identified. While the CWT program provides valuable information on exploitation rates and abundance, its spatio-temporal resolution is limited. Particularly valuable additions would be life-stage-specific estimates of mortality rates from tagging studies, as well as increased monitoring of smolt and recruit abundance to develop run reconstructions for a larger number of stocks that are independent of several of the assumptions associated with CWT recoveries. Such data are necessary to identify specific mechanisms limiting recruitment during marine residence, to disentangle changes in maturation age from late-stage marine mortality, and to better understand how productivity will be impacted by future environmental changes. These data also provide essential inputs to our second recommended process—developing life cycle models to evaluate how stock dynamics are influenced by concurrent nonstationary patterns in survival and age-at-maturity, as well as to better understand cumulative effects across life stages.

Fishery and conservation managers have specific levers that can influence population trajectories including harvest restrictions, hatchery enhancement, freshwater habitat remediation, and addressing adjacent risks (e.g., aquaculture interactions, prey availability, predator abundance). Despite the challenges of Chinook salmon management, there are recent positive examples here. For instance, many Chinook salmon populations assessed by the CTC have had above average escapement in recent years, presumably due to management decisions resulting in target exploitation rates below levels associated with maximum sustainable yield (CTC 2021*a*). Other populations have exhibited increased escapement as the relative proportion of natural spawning individuals has increased (CTC 2021*b*). Nevertheless, management interventions continue to be complicated by fundamental differences among stakeholders, generally resulting from competing trade-offs among user groups. Life cycle models provide an opportunity to resolve these issues transparently. Management strategy evaluations link models of population dynamics (including but not limited to life cycle models) to simulated management actions to quantify the benefits of specific interventions while accounting for uncertainty (Punt et al. 2016). We recognize that such investments into science-based management may be costly and prolonged, but we believe they can provide an efficient and adaptable (in light of continued scientific observations and hypotheses) framework to address widespread issues, such as the poor survival and declining age-at-maturity of Chinook salmon demonstrated here.

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Data availability

The data and code underpinning these analyses are available from https://github.com/CamFreshwater/chinDyn and archived with Zenodo at https://doi.org/10.5281/zenodo.6473951.

Author information

Competing interests

The authors declare there are no competing interests.

Supplementary material

Supplementary data are available with the article at https://doi.org/10.1139/cjfas-2021-0312.

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